In [1]:
```python
import pandas as pd
data=pd.read_csv("C:/Users/HP/Downloads/train (1).csv")
```

In [2]:
```python
data
```

Out[2]:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	Alley	LotShape	LandContour	Utilities	LotConfig
0	1	60.0	RL	65.0	8450.0	Pave	NaN	Reg	Lvl	AllPub	Inside
1	2	20.0	RL	80.0	9600.0	Pave	NaN	Reg	Lvl	AllPub	FR2
2	3	60.0	RL	68.0	11250.0	Pave	NaN	IR1	Lvl	AllPub	Inside
3	4	70.0	RL	60.0	9550.0	Pave	NaN	IR1	Lvl	AllPub	Corner
4	5	60.0	RL	84.0	14260.0	Pave	NaN	IR1	Lvl	AllPub	FR2
...	...	...	...	...	...	...	...	...	...	...	...
1455	1456	60.0	RL	62.0	7917.0	Pave	NaN	Reg	Lvl	AllPub	Inside
1456	1457	20.0	RL	85.0	13175.0	Pave	NaN	Reg	Lvl	AllPub	Inside
1457	1458	70.0	RL	66.0	9042.0	Pave	NaN	Reg	Lvl	AllPub	Inside
1458	1459	20.0	RL	68.0	9717.0	Pave	NaN	Reg	Lvl	AllPub	Inside
1459	1460	20.0	RL	75.0	9937.0	Pave	NaN	Reg	Lvl	AllPub	Inside

1460 rows × 11 columns

In [3]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 11 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Id           1460 non-null   int64  
 1   MSSubClass   1436 non-null   float64
 2   MSZoning     1448 non-null   object 
 3   LotFrontage  1201 non-null   float64
 4   LotArea      1454 non-null   float64
 5   Street       1460 non-null   object 
 6   Alley        91 non-null     object 
 7   LotShape     1460 non-null   object 
 8   LandContour  1460 non-null   object 
 9   Utilities    1460 non-null   object 
 10  LotConfig    1460 non-null   object 
dtypes: float64(3), int64(1), object(7)
memory usage: 125.6+ KB
```

In [4]:
```
data.isnull().sum()
```

Out[4]:
```
Id                0
MSSubClass       24
MSZoning         12
LotFrontage     259
LotArea           6
Street            0
Alley          1369
LotShape          0
LandContour       0
Utilities         0
LotConfig         0
dtype: int64
```

In [5]:
```
round(data.isnull().mean()*100,1)
```

Out[5]:
```
Id              0.0
MSSubClass      1.6
MSZoning        0.8
LotFrontage    17.7
LotArea         0.4
Street          0.0
Alley          93.8
LotShape        0.0
LandContour     0.0
Utilities       0.0
LotConfig       0.0
dtype: float64
```

In [6]:
```
data.drop(columns="Alley",inplace=True)
```

In [7]:
```
data
```

Out[7]:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	LotShape	LandContour	Utilities	LotConfig
0	1	60.0	RL	65.0	8450.0	Pave	Reg	Lvl	AllPub	Inside
1	2	20.0	RL	80.0	9600.0	Pave	Reg	Lvl	AllPub	FR2
2	3	60.0	RL	68.0	11250.0	Pave	IR1	Lvl	AllPub	Inside
3	4	70.0	RL	60.0	9550.0	Pave	IR1	Lvl	AllPub	Corner
4	5	60.0	RL	84.0	14260.0	Pave	IR1	Lvl	AllPub	FR2
...	...	...	...	...	...	...	...	...	...	...
1455	1456	60.0	RL	62.0	7917.0	Pave	Reg	Lvl	AllPub	Inside
1456	1457	20.0	RL	85.0	13175.0	Pave	Reg	Lvl	AllPub	Inside
1457	1458	70.0	RL	66.0	9042.0	Pave	Reg	Lvl	AllPub	Inside
1458	1459	20.0	RL	68.0	9717.0	Pave	Reg	Lvl	AllPub	Inside
1459	1460	20.0	RL	75.0	9937.0	Pave	Reg	Lvl	AllPub	Inside

1460 rows × 10 columns

In [8]:
```
#data.drop("Alley",axis=1) #columns
```

In [9]:
```
#eg

data.drop([0, 1], axis=0) # rows
```

Out[9]:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	LotShape	LandContour	Utilities	LotConfig
2	3	60.0	RL	68.0	11250.0	Pave	IR1	Lvl	AllPub	Inside
3	4	70.0	RL	60.0	9550.0	Pave	IR1	Lvl	AllPub	Corner
4	5	60.0	RL	84.0	14260.0	Pave	IR1	Lvl	AllPub	FR2
5	6	50.0	RL	85.0	14115.0	Pave	IR1	Lvl	AllPub	Inside
6	7	20.0	RL	75.0	10084.0	Pave	Reg	Lvl	AllPub	Inside
...	...	...	...	...	...	...	...	...	...	...
1455	1456	60.0	RL	62.0	7917.0	Pave	Reg	Lvl	AllPub	Inside
1456	1457	20.0	RL	85.0	13175.0	Pave	Reg	Lvl	AllPub	Inside
1457	1458	70.0	RL	66.0	9042.0	Pave	Reg	Lvl	AllPub	Inside
1458	1459	20.0	RL	68.0	9717.0	Pave	Reg	Lvl	AllPub	Inside
1459	1460	20.0	RL	75.0	9937.0	Pave	Reg	Lvl	AllPub	Inside

1458 rows × 10 columns

In [10]:
```
data
```

Out[10]:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	LotShape	LandContour	Utilities	LotConfig
0	1	60.0	RL	65.0	8450.0	Pave	Reg	Lvl	AllPub	Inside
1	2	20.0	RL	80.0	9600.0	Pave	Reg	Lvl	AllPub	FR2
2	3	60.0	RL	68.0	11250.0	Pave	IR1	Lvl	AllPub	Inside
3	4	70.0	RL	60.0	9550.0	Pave	IR1	Lvl	AllPub	Corner
4	5	60.0	RL	84.0	14260.0	Pave	IR1	Lvl	AllPub	FR2
...	...	...	...	...	...	...	...	...	...	...
1455	1456	60.0	RL	62.0	7917.0	Pave	Reg	Lvl	AllPub	Inside
1456	1457	20.0	RL	85.0	13175.0	Pave	Reg	Lvl	AllPub	Inside
1457	1458	70.0	RL	66.0	9042.0	Pave	Reg	Lvl	AllPub	Inside
1458	1459	20.0	RL	68.0	9717.0	Pave	Reg	Lvl	AllPub	Inside
1459	1460	20.0	RL	75.0	9937.0	Pave	Reg	Lvl	AllPub	Inside

1460 rows × 10 columns

In [12]:
```
#data=data.drop("Alley",axis=1)
```

In [13]:
```
data
```
Out[13]:

	Id	MSSubClass	MSZoning	LotFrontage	LotArea	Street	LotShape	LandContour	Utilities	LotConfig
0	1	60.0	RL	65.0	8450.0	Pave	Reg	Lvl	AllPub	Inside
1	2	20.0	RL	80.0	9600.0	Pave	Reg	Lvl	AllPub	FR2
2	3	60.0	RL	68.0	11250.0	Pave	IR1	Lvl	AllPub	Inside
3	4	70.0	RL	60.0	9550.0	Pave	IR1	Lvl	AllPub	Corner
4	5	60.0	RL	84.0	14260.0	Pave	IR1	Lvl	AllPub	FR2
...	...	...	...	...	...	...	...	...	...	...
1455	1456	60.0	RL	62.0	7917.0	Pave	Reg	Lvl	AllPub	Inside
1456	1457	20.0	RL	85.0	13175.0	Pave	Reg	Lvl	AllPub	Inside
1457	1458	70.0	RL	66.0	9042.0	Pave	Reg	Lvl	AllPub	Inside
1458	1459	20.0	RL	68.0	9717.0	Pave	Reg	Lvl	AllPub	Inside
1459	1460	20.0	RL	75.0	9937.0	Pave	Reg	Lvl	AllPub	Inside

1460 rows × 10 columns

In [14]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Id           1460 non-null   int64  
 1   MSSubClass   1436 non-null   float64
 2   MSZoning     1448 non-null   object 
 3   LotFrontage  1201 non-null   float64
 4   LotArea      1454 non-null   float64
 5   Street       1460 non-null   object 
 6   LotShape     1460 non-null   object 
 7   LandContour  1460 non-null   object 
 8   Utilities    1460 non-null   object 
 9   LotConfig    1460 non-null   object 
dtypes: float64(3), int64(1), object(6)
memory usage: 114.2+ KB
```

In [15]:
```
numeric=data.select_dtypes("number")
```

In [16]:
```
numeric
```

Out[16]:

	Id	MSSubClass	LotFrontage	LotArea
0	1	60.0	65.0	8450.0
1	2	20.0	80.0	9600.0
2	3	60.0	68.0	11250.0
3	4	70.0	60.0	9550.0
4	5	60.0	84.0	14260.0
...	...	...	...	...
1455	1456	60.0	62.0	7917.0
1456	1457	20.0	85.0	13175.0
1457	1458	70.0	66.0	9042.0
1458	1459	20.0	68.0	9717.0
1459	1460	20.0	75.0	9937.0

1460 rows × 4 columns

In [17]:
```
numeric.isnull().sum()
```

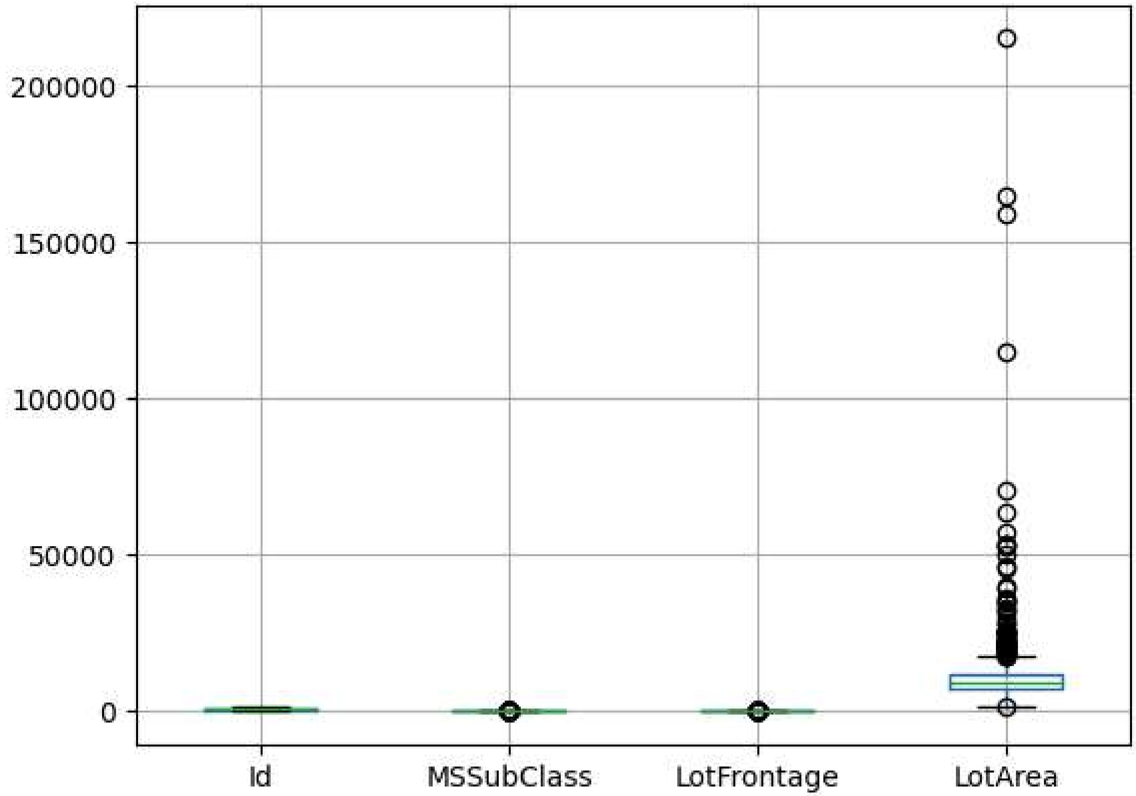
Out[17]:
```
Id               0
MSSubClass      24
LotFrontage    259
LotArea          6
dtype: int64
```

In [18]:
```
numeric.describe()
```

Out[18]:

	Id	MSSubClass	LotFrontage	LotArea
count	1460.000000	1436.000000	1201.000000	1454.000000
mean	730.500000	56.928969	70.049958	10524.268913
std	421.610009	42.355364	24.284752	10000.607481
min	1.000000	20.000000	21.000000	1300.000000
25%	365.750000	20.000000	59.000000	7558.500000
50%	730.500000	50.000000	69.000000	9485.000000
75%	1095.250000	70.000000	80.000000	11613.500000
max	1460.000000	190.000000	313.000000	215245.000000

In [19]:
```
numeric.boxplot()
```
Out[19]: <AxesSubplot:>



median-positional avg- not sensitive to outliers-(replace with median)

In [ ]:

In [20]:
```
numeric.describe()
```

Out[20]:

	Id	MSSubClass	LotFrontage	LotArea
count	1460.000000	1436.000000	1201.000000	1454.000000
mean	730.500000	56.928969	70.049958	10524.268913
std	421.610009	42.355364	24.284752	10000.607481
min	1.000000	20.000000	21.000000	1300.000000
25%	365.750000	20.000000	59.000000	7558.500000
50%	730.500000	50.000000	69.000000	9485.000000
75%	1095.250000	70.000000	80.000000	11613.500000
max	1460.000000	190.000000	313.000000	215245.000000

In [21]:
```
for i in numeric:
    sns.displot(numeric[i])
    plt.show()
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_10984\3884094485.py in <module>
      1 for i in numeric:
----> 2     sns.displot(numeric[i])
      3     plt.show()

NameError: name 'sns' is not defined
```

In [ ]:
```
for col in numeric:
    sns.displot(numeric[col], kde=True)
    plt.title(col)
    plt.xlabel(col)
    plt.ylabel('Frequency')
    plt.show()
```

In [ ]:
```
numeric.skew()
```

In [22]:
```
numeric.isnull().sum()
```

Out[22]:
```
Id               0
MSSubClass      24
LotFrontage    259
LotArea          6
dtype: int64
```

In [23]:
```
med=numeric["LotArea"].median()
med
```

Out[23]:
```
9485.0
```

In [24]:
```
numeric["LotArea"].fillna(med,inplace=True)
```

In [25]:
```
numeric.isnull().sum()
```

Out[25]:
```
Id               0
MSSubClass      24
LotFrontage    259
LotArea          0
dtype: int64
```

In [26]:
```
x=[1,2,3,4,5,na,na,na,na,na]
x=3

x=[1,2,3,4,5,3,3,3,3,3]
x=[1,2,3,4,5,3,1,2,4,5]
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_10984\2333363417.py in <module>
----> 1 x=[1,2,3,4,5,na,na,na,na,na]
      2 x=3
      3 
      4 x=[1,2,3,4,5,3,3,3,3,3]
      5 x=[1,2,3,4,5,3,1,2,4,5]

NameError: name 'na' is not defined
```

lets consider lotfrontage column , it has 18% data missing with skew 2.16, since we can replace with median , 216 values will be imputed by the same value, ie: the same value repeated many time, so we can do another methos

In [27]:
```
from sklearn.experimental import enable_iterative_imputer
from sklearn.impute import IterativeImputer
```

In [28]:
```python
# Initialize the IterativeImputer
imputer = IterativeImputer()

# Fit and transform the data
numeric["LotFrontage"] = imputer.fit_transform(numeric[["LotFrontage"]])

print(numeric)
```

```
        Id  MSSubClass  LotFrontage  LotArea
0        1        60.0         65.0   8450.0
1        2        20.0         80.0   9600.0
2        3        60.0         68.0  11250.0
3        4        70.0         60.0   9550.0
4        5        60.0         84.0  14260.0
...    ...         ...          ...      ...
1455  1456        60.0         62.0   7917.0
1456  1457        20.0         85.0  13175.0
1457  1458        70.0         66.0   9042.0
1458  1459        20.0         68.0   9717.0
1459  1460        20.0         75.0   9937.0

[1460 rows x 4 columns]
```

In [ ]:

In [29]:
```python
numeric.isnull().sum()
```

Out[29]:
```
Id              0
MSSubClass     24
LotFrontage     0
LotArea         0
dtype: int64
```

In [30]:
```python
numeric.skew()
```

Out[30]:
```
Id              0.000000
MSSubClass      1.405217
LotFrontage     2.384950
LotArea        12.210558
dtype: float64
```

your job 😊

MSSubClass replace by mean

In [31]:
```
data.isnull().sum()
```

Out[31]:
```
Id               0
MSSubClass      24
MSZoning        12
LotFrontage    259
LotArea          6
Street           0
LotShape         0
LandContour      0
Utilities        0
LotConfig        0
dtype: int64
```

In [32]:
```
data.update(numeric)
```

In [33]:
```
data.isnull().sum()
```

Out[33]:
```
Id              0
MSSubClass     24
MSZoning       12
LotFrontage     0
LotArea         0
Street          0
LotShape        0
LandContour     0
Utilities       0
LotConfig       0
dtype: int64
```

In [34]:
```
cat=data.select_dtypes("object")
cat
```

Out[34]:

	MSZoning	Street	LotShape	LandContour	Utilities	LotConfig
0	RL	Pave	Reg	Lvl	AllPub	Inside
1	RL	Pave	Reg	Lvl	AllPub	FR2
2	RL	Pave	IR1	Lvl	AllPub	Inside
3	RL	Pave	IR1	Lvl	AllPub	Corner
4	RL	Pave	IR1	Lvl	AllPub	FR2
...	...	...	...	...	...	...
1455	RL	Pave	Reg	Lvl	AllPub	Inside
1456	RL	Pave	Reg	Lvl	AllPub	Inside
1457	RL	Pave	Reg	Lvl	AllPub	Inside
1458	RL	Pave	Reg	Lvl	AllPub	Inside
1459	RL	Pave	Reg	Lvl	AllPub	Inside

1460 rows × 6 columns

In [35]:
```
cat["MSZoning"].value_counts()
```

Out[35]:
```
RL         1141
RM          216
FV           65
RH           16
C (all)      10
Name: MSZoning, dtype: int64
```

In [36]:
```
cat["MSZoning"].mode()
```

Out[36]:
```
0    RL
Name: MSZoning, dtype: object
```

In [37]:
```
cat.isnull().sum()
```

Out[37]:
```
MSZoning       12
Street          0
LotShape        0
LandContour     0
Utilities       0
LotConfig       0
dtype: int64
```

In [38]:
```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 1460 entries, 0 to 1459
Data columns (total 10 columns):
 #   Column       Non-Null Count  Dtype  
---  ------       --------------  -----  
 0   Id           1460 non-null   int64  
 1   MSSubClass   1436 non-null   float64
 2   MSZoning     1448 non-null   object 
 3   LotFrontage  1460 non-null   float64
 4   LotArea      1460 non-null   float64
 5   Street       1460 non-null   object 
 6   LotShape     1460 non-null   object 
 7   LandContour  1460 non-null   object 
 8   Utilities    1460 non-null   object 
 9   LotConfig    1460 non-null   object 
dtypes: float64(3), int64(1), object(6)
memory usage: 114.2+ KB
```

In [39]:
```
X=cat["MSZoning"].mode()
X
```

Out[39]:
```
0    RL
Name: MSZoning, dtype: object
```

In [58]:
```
X[0]
```

Out[58]:
```
'RL'
```

In [54]:
```
cat["MSZoning"].fillna(X[0],inplace=True)
```

In [55]:
```
# your job:  replace MSZoning with mode


cat["MSZoning"].isnull().sum()
```

Out[55]:
```
0
```

In [49]:
```
# plan b
```

In [43]:
```
import pandas as pd
import numpy as np



# Function for random imputation
def random_imputation(column):
    # Get non-missing values
    non_missing = column.dropna()

    # Generate random values based on the distribution of non-missing values
    random_values = np.random.choice(non_missing)

    # Combine original and random values
    column[column.isnull()] = random_values

    return column

# Apply random imputation to each numeric column
for col in data.select_dtypes(include='object'):
    data[col] = random_imputation(data[col])

print(data)
```

```
        Id  MSSubClass MSZoning  LotFrontage  LotArea Street LotShape  \
0        1        60.0       RL         65.0   8450.0   Pave      Reg   
1        2        20.0       RL         80.0   9600.0   Pave      Reg   
2        3        60.0       RL         68.0  11250.0   Pave      IR1   
3        4        70.0       RL         60.0   9550.0   Pave      IR1   
4        5        60.0       RL         84.0  14260.0   Pave      IR1   
...    ...         ...      ...          ...      ...    ...      ...   
1455  1456        60.0       RL         62.0   7917.0   Pave      Reg   
1456  1457        20.0       RL         85.0  13175.0   Pave      Reg   
1457  1458        70.0       RL         66.0   9042.0   Pave      Reg   
1458  1459        20.0       RL         68.0   9717.0   Pave      Reg   
1459  1460        20.0       RL         75.0   9937.0   Pave      Reg   

     LandContour Utilities LotConfig  
0            Lvl    AllPub    Inside  
1            Lvl    AllPub       FR2  
2            Lvl    AllPub    Inside  
3            Lvl    AllPub    Corner  
4            Lvl    AllPub       FR2  
...          ...       ...       ...  
1455         Lvl    AllPub    Inside  
1456         Lvl    AllPub    Inside  
1457         Lvl    AllPub    Inside  
1458         Lvl    AllPub    Inside  
1459         Lvl    AllPub    Inside  

[1460 rows x 10 columns]

C:\Users\HP\AppData\Local\Temp\ipykernel_10984\1085257359.py:15: SettingWithCopyWarning: 
A value is trying to be set on a copy of a slice from a DataFrame

See the caveats in the documentation: https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy (https://pandas.pydata.org/pandas-docs/stable/user_guide/indexing.html#returning-a-view-versus-a-copy)
  column[column.isnull()] = random_values
```

In [44]:
```
data.isnull().sum()
```

Out[44]:
```
Id              0
MSSubClass     24
MSZoning        0
LotFrontage     0
LotArea         0
Street          0
LotShape        0
LandContour     0
Utilities       0
LotConfig       0
dtype: int64
```

In [ ]:
```
```

In [ ]:
```
```

In [ ]:
```
```